

**Time:** 5 minutes**Outcome:** Explain Consumer Surplus & Willingness to Pay.**Difficulty:**

●●○ Intermediate

## Consumer Surplus & Willingness to Pay



### THE CORE IDEA

Consumer surplus is the difference between a buyer's willingness to pay and the price actually paid. Surplus analysis compares value created with the opportunity cost of the resources used. Demand can be interpreted as buyers' marginal willingness to pay. The gain from an additional trade is the buyer's willingness to pay minus the seller's cost, not either side's individual surplus alone.



### HOW TO RECOGNIZE IT

- Look for a buyer's willingness to pay and the market price actually paid.
- Individual consumer surplus equals willingness to pay minus price.
- On a graph, consumer surplus is the area below demand and above the market price up to the quantity traded.
- A buyer whose willingness to pay is below price does not purchase and earns no consumer surplus.

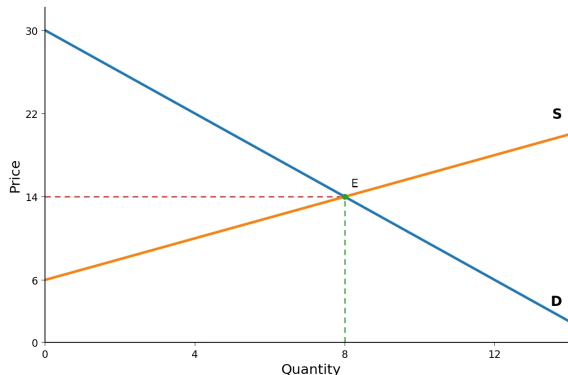


### WATCH OUT

Do not assume every trade creates surplus. The gain from an additional trade is the buyer's willingness to pay minus the seller's cost, not either side's individual surplus alone.



### WORKED EXAMPLE: CALCULATING CONSUMER SURPLUS



At equilibrium, price is \$14 and quantity is 8. The demand intercept of \$30 represents the highest willingness to pay. Consumer surplus is the triangle above price and below demand:  $\frac{1}{2} \times 8 \times (\$30 - \$14) = \$64$ .



### CHECK YOURSELF

Why is the rectangle below market price not consumer surplus?

**READY?**

Return to the game and master this concept.